PCA before CCA: Useful or Problematic?

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1 Introduction

Canonical Correlation Analysis (CCA) is one of the most popular methods to analyze associations between two sets of variables, provided the sets are from the same sample. However, CCA is limited at higher dimensions by the computational cost of performing CCA. In some cases, where sample size is smaller than variables, CCA may be completely impossible. To combat this, an option may be to first perform Principal Components Analysis (PCA) prior to performing CCA. However, combining these two methods may have unintended consequences. In this project, we will use a simulation study to better understand what the strengths and weaknesses of performing PCA prior to CCA are, and ultimately, if it is a good choice.

To test this, we ran a simulation study. The data is purely synthetic, as this allows us to know the actual relationships to see exactly how well CCA performs in each scenario. We simulated two groups of 25 measurements, for a total of 50 measurements. This project was influenced by the Human Connectome Project (HCP), by Smith et al. (2015) [4]. In the HCP, measurements are taken both of brain activity and behavior characteristics. The measurements in the simulation study can be thought of as those. Of the 25 measurements per group, 5 are high-variance between individuals, and the rest are low-variance between individuals. Medium-variance was not included as this data is purely simulated, and the difference between low-variance and high-variance should best illustrate the comparison.

Using this simulated data, we performed a Monte Carlo Simulation Study. The data was drawn using multivariate normal distributions. We varied both how many individuals were sampled, and also which 3 variables we chose to have some correlation (3 high variance, 3 low variance, 1 high variance 2 low variance, and 2 high variance 1 low variance). Finally, we varied how many Principal Components were kept (1, 3, 5, 10, and 15). The results show that doing PCA before CCA is functionally a wager that the high variance variables contain the relationship. Performing PCA first helps find the truth compared to just CCA when the high variance variables contain the relationship. However, when they don't, PCA is a major detriment to finding the truth. We

also observed that, with a high enough sample size, PCA with a large number of PCs was functionally indistinguishable from CCA, but in many situations, PCA with a small number of PCs was never able to come close to the truth.

2 Methods

PCA and CCA are both widely used methods in multivariate analysis. PCA revolves around the idea of Singular Value Decomposition (SVD). SVD states that any matrix $\mathbf{A}$ can be expressed as $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices, and $\mathbf{\Sigma}$ is a diagonal matrix of singular values. PCA is an application of SVD to a data matrix, $\mathbf{X}$. PCA searches for the directions of maximum variance in the data. To do this, we first center $\mathbf{X}$ by subtracting each column's mean, giving $\mathbf{X}_c$, and compute the sample covariance matrix $\widehat{\mathbf{S}} = \mathbf{X}_c^T\mathbf{X}_c/(n-1)$, where n is the number of observations. The variance of the data projected onto a unit vector $\mathbf{v}$ is $\mathbf{v}^T\widehat{\mathbf{S}}\mathbf{v}$, which is maximized by the eigenvector of $\widehat{\mathbf{S}}$ with the largest eigenvalue. The eigenvectors $\mathbf{v}_j$ are the principal directions, the directions in which variance is maximized, and the eigenvalues λ_j are the variances along them. The right singular vectors of $\mathbf{X}_c$ (the columns of $\mathbf{V}$) are the eigenvectors, with $\lambda_j = \sigma_j^2/(n-1)$. The principal components are the projected data, $\mathbf{Y}_q = \mathbf{X}_c\mathbf{V}_q = \mathbf{U}_q\mathbf{\Sigma}_q$. PCA is almost always used as a dimension reduction technique. By keeping only the first k principal components, the most important part of the data is kept, while the other $p-k$ principal components may be discarded to reduce dimensions.

On the other hand, CCA requires one dataset split into two groups. Gao described CCA as "[a method which] identifies associations between two sets of variables, where all variables are measured on the same sample." [1] So, the requirements for CCA are stricter than PCA. The nature of CCA is such that the covariance matrix must have specific blocks that are positive definite to be performed. So, in some cases, PCA may provide a way for CCA to be performed when otherwise it couldn't be. Specifically, we divide the dataset, $\mathbf{X}$, into the two groups, $\mathbf{X}^{(1)}$ and $\mathbf{X}^{(2)}$, which need not be the same size. The covariance matrices for these are $\mathbf{\Sigma}_{11}$ and $\mathbf{\Sigma}_{22}$, respectively. Both of these covariance matrices must be positive definite for CCA to be performed. The covariance between the two, $\mathrm{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)})$, is used for the other two blocks of $\mathbf{\Sigma}$, namely $\mathbf{\Sigma}_{12} = \mathbf{\Sigma}_{21}^T$. In CCA, we define $U = \mathbf{a}^T\mathbf{X}^{(1)}$ and $V = \mathbf{b}^T\mathbf{X}^{(2)}$, and look to find $\mathbf{a}, \mathbf{b}$ to maximize $\mathrm{Corr}\big(\mathbf{a}^\top\mathbf{X}^{(1)}, \mathbf{b}^\top\mathbf{X}^{(2)}\big) = \frac{\mathbf{a}^\top\mathbf{\Sigma}_{12}\mathbf{b}}{\sqrt{\mathbf{a}^\top\mathbf{\Sigma}_{11}\mathbf{a}}\sqrt{\mathbf{b}^\top\mathbf{\Sigma}_{22}\mathbf{b}}}$. To achieve this, we form $\mathbf{M} = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1/2}$. We can observe that the eigenvector corresponding to the largest eigenvalue of $\mathbf{M}\mathbf{M}^T$, $\mathbf{e}_1$, is such that $\mathbf{a}_1 = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{e}_1$. Similarly, the eigenvector corresponding to the largest eigenvalue of $\mathbf{M}^T\mathbf{M}$, $\mathbf{f}_1$, is such that $\mathbf{b}_1 = \mathbf{\Sigma}_{22}^{-1/2}\mathbf{f}_1$. $\mathbf{M}\mathbf{M}^T$ and $\mathbf{M}^T\mathbf{M}$ both have the same eigenvalues, just different eigenvectors. The square root of the largest eigenvalue of both matrices, λ_1, is the first canonical correlation, defined as ρ_1. Taking the kth largest eigenvalues

and eigenvectors provides the kth canonical correlation. ρ_k shows how strongly the two groups are related, at their kth strongest link, unrelated to the first $k-1$ strongest links. U and V are numbers describing each observation's set 1 and set 2 "scores", formed by using $\mathbf{a}$ and $\mathbf{b}$ as weights, respectively.

PCA is frequently used as a precursor to various methods, although the ramifications of doing such may be significant. PCA's ability to reduce dimensions is extremely useful, and its principal components are frequently interpretable. However, PCA does have drawbacks. As with any dimension reduction technique, PCA invariably loses some information when reducing. Thus, the choice of how many principal components to keep, k, is an extremely important choice. With too small of a k, part or all of the signal may be thrown out before even moving on to the next method, while with too large of a k, the dimension reduction may not be strong enough to perform the intended role. PCA also crucially maximizes variance, which may cause issues if that is not where the signal is. Finally, there is a loss of interpretability, as any new method is now performing on principal components instead of the original data.

To perform PCA before CCA, PCA is performed on each group of $\mathbf{X}$ separately. CCA still needs two separate groups of variables, so PCA on $\mathbf{X}$ itself would not provide a result that CCA can accept. Performing PCA before CCA is not a novel procedure. In the Human Connectome Project (HCP), as discussed by Smith et al. in 2015, the goal was to investigate the "relationship between individual subjects' functional connectomes and 280 behavioral and demographic measures [... from] non-imaging data from 461 subjects in the Human Connectome Project". They did this using a "200×200 connectome for each subject." To perform CCA, the covariance matrix is needed. Using CCA on data of this size would be extremely computationally expensive and potentially infeasible due to whether or not the sample covariance matrix is invertible. So, the authors first performed PCA and kept the first 100 principal components, and then ran CCA on that. This simulation study hopes to uncover whether there were any unintended consequences of this pipeline.

3 Data

The data was all simulated using R [2] [3]. First, the parameters were initialized. We specified the dimension of both $\mathbf{X}$s, the number of high-variance variables, the variance the high-variance variables would have, and the correlation between any two linked variables. Next, we set two vectors, the training set sizes, and the kept principal components. Finally, the number of replications was set. We chose 25 and 5 as the initial starting vectors and high-variance values for computational ease. 25 is large enough to allow for different values of principal components to keep while small enough to be able to run the simulation in minutes. The variance was chosen to be 10 after seeing clean separation from

10 after testing different numbers. The correlation between any two linked variables is set at 0.2, which was also chosen after testing different values. Values much lower than 0.2 made everything fail while values much higher than 0.2 made all methods work. For the training set sizes, we first picked 60 as it is just barely larger than the 50 variables. We incrementally scaled up to 100, 200, 500, and then 1000. 1000 was chosen as a value at which all methods show convergence. Finally, for kept PCs we kept 1, 3, 5, 10, and 15. We also will perform CCA with no PCA, which would be the same as keeping 25 PCs. With only 25 variables, keeping 15 is already more than half, and 1, 3, 5, and 10 are proportional to around 4%, 12%, 20%, 40% retention. It is important to know that since PCA works on one set at a time, there are 25 variables to consider here as opposed to the 50 discussed in CCA. So, each of these numbers is purely per set.

With the parameters initialized, the covariance matrices were built. Each matrix is built so that it can be used in a multivariate normal distribution to draw the data. We built a vector containing which variables are linked based on the scenario (3 high-variance, 3 low-variance, 2 high-variance and 1 low-variance, and 1 high-variance and 2 low-variance). The same-numbered variables in each matrix are linked. So, for example, if 1, 6, and 7 were drawn, in both sets the first, sixth, and seventh variables are linked. Next, we made a correlation matrix for the first set starting with the identity matrix and then adding the 0.2 correlation between the linked variables, denoted as $\boldsymbol{\rho}_{11}$. This matrix is duplicated for the second set to make $\boldsymbol{\rho}_{22}$. Finally, for the between-set correlation, we start with a matrix of all 0s, and replace the linked variables' entries with 0.2, forming $\boldsymbol{\rho}_{12}$. The transpose of $\boldsymbol{\rho}_{12}$ is taken, creating $\boldsymbol{\rho}_{21}$. The correlation matrix is made with $\boldsymbol{\rho}_{11}$ in the top left, $\boldsymbol{\rho}_{12}$ in the top right, $\boldsymbol{\rho}_{21}$ in the bottom left, and $\boldsymbol{\rho}_{22}$ in the bottom right, creating $\boldsymbol{\rho}$. To make the covariance matrix from the correlation matrix, we use the formula $\sigma_{ij} = \rho_{ij}\sqrt{\sigma_{ii}}\sqrt{\sigma_{jj}}$. So, using the variances defined in the parameters, we make a vector with 25 terms, repeating 1s. We replace the 1s with the variance of 10 defined in the parameters section for the high-variance variables, making a vector of 20 1s and 5 10s. We then create matrix $\mathbf{V}$ from this vector such that the vector is the diagonal of matrix $\mathbf{V}$, and all non-diagonal entries are 0. With this, we can take $\mathbf{V}^{1/2}\boldsymbol{\rho}\mathbf{V}^{1/2}$, creating covariance matrix $\boldsymbol{\Sigma}$.

Before generating data, we also calculate the truth by performing CCA on the population, rather than the sample. This gives us the correct answers to compare to when doing the simulation study. R's `cancor` function requires data, so we define a CCA function for the first pair manually. Breaking $\boldsymbol{\Sigma}$ into the same blocks from $\boldsymbol{\rho}$ allows for CCA to be performed. To do this, we compute $\mathbf{M} = \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}$, and compute $\mathbf{M}\mathbf{M}^T$. We then take the top eigenvalue of $\mathbf{M}\mathbf{M}^T$, λ_1. We finally can compute the first canonical correlation as $\sqrt{\lambda_1} = \rho_1^*$.

To make the data, we use the `MASS` package in R. We first create a $\boldsymbol{\mu}$ vector of 0s with length 50. We then call the `mvrnorm` command, using $\boldsymbol{\mu}$ as the mean

and $\mathbf{\Sigma}$ as the variance. We finally create a simulation function to replicate this call $n = 5000$ times. Figure 1 provides a look at one sample's true correlation, sample correlation, and sample variance.

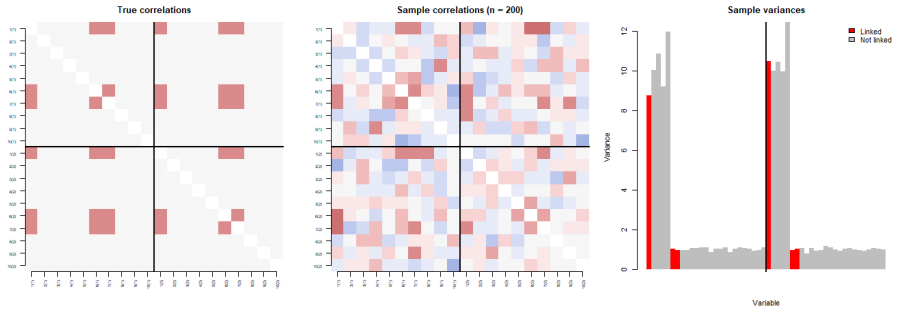


Figure 1: True Correlations, Sample Correlations, Sample Variances of a sample with one High-Variance variable and two Low-Variance variables

Both a training set and a test set are drawn from this distribution. The methods are fit to the training data, and then scored based on the test set. The test set was made with 5000 observations to be large enough to combat any randomness in set generation. As discussed in the parameter section, the training set varies from 60 observations to 1000 observations.

We expect to see unreliable estimations at lower training set sizes. We also expect to see unreliable estimations (likely close to 0) even at higher training set sizes, when the linked variables are low-variance and we keep few principal components.

4 Results

CCA, and CCA with PCA keeping k principal components, using each of the four scenarios (3 High Variance, 2 High Variance and 1 Low Variance, 1 High Variance and 2 Low Variance, and 3 Low Variance) were evaluated with 3 metrics. Firstly, we looked to see how correlated each method showed U and V to be on the training data. Secondly, we looked to see how correlated each method showed U and V to be on the test data. Finally, given that we have the true weights, we checked how close each method was to recovering the true U. Notably, since both $\mathbf{X}^{(1)}$ and $\mathbf{X}^{(2)}$ were built the same way, checking for the correct V would be expected to be redundant.

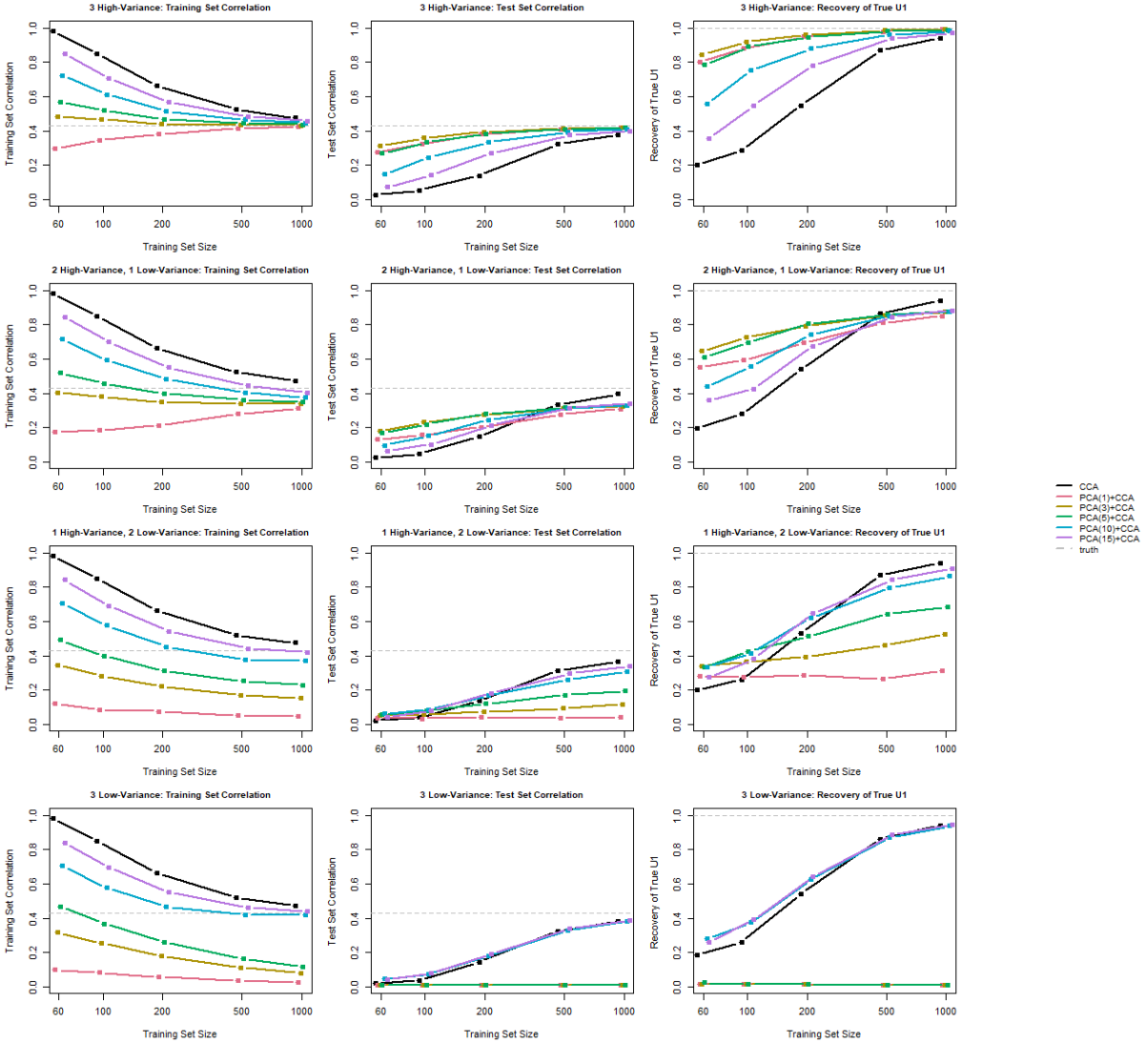


Figure 2: Left Column: Results of each scenario on the training set; Middle Column: Results of each scenario on the test set; Right Column: Results of each scenario on recovery of U_1

Figure 2 shows the results of all tests. The first column shows the results of the training set tests. It is immediately apparent, especially at low sample sizes, that the more principal components are kept, the larger the training set correlation, with CCA (the same as keeping all principal components) the highest. This follows from the way that this workflow operates. Since CCA searches to maximize correlation, it searches all of the data to find the highest corre-

lation between variables. So, the fewer principal components kept, the fewer variables exist to search for correlation. At smaller n, there is a high risk of overfitting. It is much more likely with a small training set that there will be a high correlation by chance. With 3 High Variance linked variables, by a training set size of 1000, all methods end up with almost the exact correlation. In fact, CCA performs arguably the worst here, being the farthest away from the true correlation until all methods produced virtually the exact same response. However, with 1 and 0 high-variance variables, we can distinctly see that only by keeping 10 or more PCs can we come close to finding the truth. The other 3 lines trend towards 0 in both scenarios. This is due to the fact that keeping fewer PCs threw away the signal, and with higher training set size, the probability of correlation by chance trends towards 0. So, performing PCA before CCA when the signal exists in lower variance variables may end up not working at all.

A similar pattern exists in the 2nd column, the test set correlation. The results we get tell a similar story as before, although even more marked. With 3 high variance variables, CCA arguably performs the worst out of all methods, although by a training set of 1000, the methods are close to indistinguishable. However, even by moving to the 2 high variance, 1 low variance scenario, CCA clearly outperforms the other methods, and by 1 high variance and 0 high variance, only when keeping 10 or more PCs is the signal found.

The same pattern shows up even when recovering the true U_1. While with all 3 variables high variance, CCA may even perform the worst, it is clearly the best with 1 or more low variance variables. Again, by the time that there are more low-variance variables than high-variance variables, any PCA with fewer than 10 principal components kept becomes useless.

Using a table, we can attempt to find the best method for each training set size. Looking at training set sizes of 60, 200, and 1000, we can derive a table, averaging over all possible scenarios. This is the best way to mimic a real situation, as in a real situation, we don't know where the signal is. So, the table below shows what method to select given relative training size.

Table 1: Mean training correlation, test correlation, and recovery of the true U_1, averaged over all scenarios. (Best Method in bold)

	n = 60			n = 200			n = 1000		
Method	Training	Test	Recovery	Training	Test	Recovery	Training	Test	Recovery
CCA	0.981	0.023	0.195	0.664	0.141	0.541	0.474	**0.380**	**0.942**
PCA(1)+CCA	0.171	0.113	0.413	0.181	0.159	0.486	0.201	0.194	0.541
PCA(3)+CCA	**0.388**	**0.138**	**0.462**	0.296	0.187	0.539	0.251	0.218	0.600
PCA(5)+CCA	0.512	0.125	0.437	0.359	0.197	0.569	0.283	0.237	0.640
PCA(10)+CCA	0.714	0.088	0.402	**0.479**	**0.231**	**0.717**	0.402	0.356	0.914
PCA(15)+CCA	0.843	0.055	0.311	0.552	0.214	0.686	**0.430**	0.366	0.926

We define best differently for each metric. For training set, the best is the closest to the true canonical correlation of 0.429 (true across all scenarios). A number notably higher than that canonical correlation is likely overfitting. For test set, we select the highest correlation possible (which also should be closest to the canonical correlation). For recovery, we try to get as close as possible to 1, as all of the weight should sum up to 1. We can see that CCA is the best when the training set gets largest, but surprisingly, the best method for a small sample set is PCA with 3 principal components. For a medium training set size of 200, the best is PCA with 10 principal components, although CCA is somewhat close. However, 10 principal components here means keeping 40% of the data, which may not be feasible with larger data matrices.

5 Discussion

This study illustrates that running PCA before CCA serves as a sort of bet that the signal in the data exists mostly or almost completely in higher variance variables. In these situations, running PCA may not only improve computation time, but also provide slightly better results. However, when the signal exists in large part in lower variance variables, running PCA may throw out the signal rather than amplifying it.

This simulation study should be taken as a toy example and has plenty of room for extension. Firstly, using synthetic data rather than real data means this data was specifically cherry-picked for this example. The idea that some variables are high variance while the rest are low variance while completely disregarding the possibility for medium variance variables is unrealistic, but provided interpretable results. The kept PCs choices were arbitrary and, noticing the gap between 5 and 10, perhaps not the ideal choices. The linked variables only had correlation of 0.2, and this value could be varied for different results. We also ran with the lowest training set at 60, which was chosen so that CCA would still run. A smaller training set size could potentially still be used alongside PCA, mirroring what Smith et al. did. This study is intended to be illustrative and exploratory, and further extensions are necessary to truly understand the impact of running PCA before CCA.

References

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